

BHARATHKUMAR RAJAGOPAL

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bharathkumar-rajagopal.github.io

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SKILLS

- [1] **Molecular Biology:** q-PCR, SDS-PAGE, Western Blotting, DNA & RNA Sequencing, Immunoelectrophoresis
- [2] **Microbiology:** Bacterial isolation, microbial culturing, MIC, Time-kill kinetics, R-based microbial diversity analysis
- [3] **Toxicology:** *In vitro* Culture, Handling *in vivo* model, Cytotoxicity Assays, Genotoxicity (Comet) Assay
- [4] **Analytical Techniques:** Confocal microscopy, SEM, Flow cytometry (FACS), HPLC, GC-MS/MS, DLS, NTA, UV-Vis, FTIR
- [5] **Bioinformatics:** Library Preparation, SWISS-MODEL, Mega X, PyMol, Mothur, QIIME 2
- [6] **Language:** Python (Biopython), R, L^AT_EX

RESEARCH EXPERIENCE

Independent Research Assistant - Unpaid | Under Scholarly Mentorship

Jul 2024 – Aug 2025

SKILLS: Microbiology, Toxicology, Mothur, R, Manuscript Preparation

- Research in lake microbiome and ecotoxicology under Dr. Boobal Rangaswamy & Contribution to scientific manuscripts for peer-reviewed publication.

Dissertation Trainee | CSIR-IITR, Lucknow

Jan 2024 – May 2024

SKILLS: Microscopic Analysis, FACS, SEM, Mammalian Cell Culture, Handling Murine Model (*in vivo*)

- Conducted *in vitro* toxicity assessment of polystyrene nanoplastics on mammalian lung fibroblast cells
- Characterised nanoplastics using DLS and NTA; performed cytotoxicity assessment (MTT, LDH, NRU, Trypan Blue Dye Exclusion, and ROS generation assays), and Genotoxicity assessment (Comet assay). BALF assessment in murine model (*in vivo*)
- Analysed data via GraphPad; identified ROS-mediated toxicity pathways

Postgraduate Research Assistant | Amity University

Jul 2023 – Dec 2023

SKILLS: Microbial Culture Establishment & Maintenance, Bioremediation, Nano-delivery systems

- Developed pesticide-degrading microbial consortia microbeads.
- Optimised bacterial release efficiency for environmental remediation.

Summer Research Intern | Centre for Bioscience and Nanoscience Research

May 2023 – Jun 2023

SKILLS: Microscopy, SEM, UV-Vis Spectroscopy, Mammalian Cell Culture, Nanoparticle Synthesis & Characterisation, Antimicrobial & Anticancer Assays

- Isolated Bacteria *Priestia megaterium* and sequenced (GenBank: OR234764.1)
- Biogenically synthesised silver nanoparticles (AgNPs) & Characterised nanoparticles via UV-Vis, FTIR, and SEM
- Evaluated antimicrobial (MIC, Time kill assay, Antimicrobial textile test) & anticancer (Hep-G2) properties

EDUCATION

Amity University, Noida | MSc Biotechnology | First Class with Distinction | (≥97th percentile in class)

2022 – 2024

Advanced Molecular Biology, Advanced Microbiology, Advanced rDNA Technology, Animal Biotechnology, Python etc.,

PSG College of Arts & Science, Coimbatore | BSc Biotechnology | First Class

2019 – 2022

- Microbiology, Cell Biology, Chemistry of Biomolecules, Enzymology, Bioinformatics, Analytical Techniques etc.,

PUBLICATIONS

- [1] Sakthivel, I., Rangaswamy, B., **Rajagopal, B.**, & Shanmugam, L. (2025). Integrating kinetic models, gene circuits, and biofilm dynamics for enhanced exopolysaccharide production in nitrifying bacterial consortia. *Journal of Microbiological Methods*, 107237. doi:10.1016/j.mimet.2025.107237
- [2] Kongaserry, A., Shanmugam, L., **Rajagopal, B.**, & Rangaswamy, B. (2025). Development of *Artocarpus heterophyllus* Lam. nanosuspension. In Review - Nanotechnology for Environmental Engineering.
- [3] **Rajagopal, B.**, Velusamy A., & Ragunathan, R. Biogenic synthesis of silver nanoparticles using endophytic bacteria from *Andrographis paniculata* - antibacterial and anticancer properties. In preparation.
- [4] Balakrishnan S., **Rajagopal, B.**[§], Velusamy A.[§], & Rangaswamy, B. Functional Dynamics Revealing Metacommunity and Organization of Microbial Populations in the Lentic Environment of Singanallur Lake. In preparation.

§ - equal contribution

CERTIFICATES

- [1] Introduction to Professional Scientific Communication | NPTEL Jan 2024 – Feb 2024
- [2] Certificate Course on Bionanotechnology | Association of Indian Biologists May 2020
- [3] Certificate course in Bioinformatics | PSGCAS Sep 2019 – May 2020
- [4] Hands-on Training: HPLC & GC-MS/MS | ICFRE-IFGTB Feb 2020
- [5] Human Molecular Genetics | NPTEL Jan 2020 – Feb 2020

REFEREES

- [1] **Dr. Boobal Rangaswamy** – Assistant Professor, PSG College of Arts & Science | boobal@psgcas.ac.in
- [2] **Dr. Shashi Sharma** – Assistant Professor, Amity University | ssharma@amity.edu
- [3] **Dr. Ravi Kant Singh** – Professor, Amity University | rksingh1@amity.edu